Bilkent University
Computer Engineering Department
CS 491 Senior Design Project I

Assessment of Innovation Form

ID and Title of the Project: ZeroTrust

Team Member IDs: 22302443 - 22201022 - 22201366 - 22303555 - 22301544

Name of the Supervisor: Mohammad Mahmoody

Below questions needs to be filled by the Innovation Expert

*Considering the definition of innovation as "**creation of better or more effective products, systems, services, or technologies that have the potential to be accepted by markets, governments, and society**", please assess the innovativeness of the design project with below questions.*

1) The project and subject that is proposed and presented to you is:

Criteria	Poor	Unsatisfactory	Satisfactory	Good	Outstanding
Technically feasible:	☐	☐	☐	☒	☐
You see enough market demand:	☐	☐	☒	☐	☐
Development planning is done well	☐	☐	☐	☐	☒
Enough research done for marketplace and competitors	☐	☐	☐	☒	☐
Delivers enough value or solves a real problem	☐	☐	☐	☒	☐

2) What is the nature of the innovation you see in this project? Do you have any suggestion to improve converting the knowledge and idea of the subject more into benefit and value?

The main innovation of this project is combining several existing technologies into a single system for secure communication and file sharing over untrusted local networks. The individual techniques are not new, but their integration into a system that does not require an internet connection, central server, or trusted network can provide useful final product.I think the project has a good technical foundation and can offer practical value, especially for organizations working in isolated or restricted places. I suggest the team focus on a few specific use cases at the end where they can show clear advantages over existing solutions like Briar and Jami.

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2) Are there any unaddressed risks that team members need to consider during implementation?

The main risk I see is the complexity of the system. Secure communication, key management, file synchronization, routing, etc. are all challenges to address. Trying to implement all of them within the project's required timeline may be tough to make. I suggest that the team separate the essential features from optional ones and first complete a working and tested core system. Another issue to consider I believe is the weak anonymity provided by onion routing in small networks. These limitations should be clearly documented and tested if possible.

3) Any suggestions while shaping go to market strategy?

I also think, like the team, that the project would be more suitable for organizations with specific security and connectivity requirements than general consumers. Possible areas include defense, emergency response, and organizations operating in environments where Internet connectivity or external cloud services cannot be guaranteed.

The initial goal should be to demonstrate a reliable product for a specific use case, rather than trying to address several different markets at the same time.

Innovation Expert

Name: Mohammad Mahmoody

Date: 21 / Oct / 2026

Signature: Mahmoody

Final Expert Score: 4 (out of 5)

Scale:
(1) Poor, (2) Unsatisfactory, (3) Satisfactory, (4) Good, (5) Outstanding